Factors Associated with Type 2 Diabetes in NHANES 2017–2018

A Real-World Evidence Analysis Using Logistic Regression

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Research Question and Objective

Research question
Which demographic and clinical factors are associated with prevalent Type 2 diabetes among adults in the United States?

Objective
To conduct a real-world evidence analysis using NHANES 2017–2018 data and estimate associations between diabetes status and selected clinical/demographic predictors.

Primary outcome

- Self-reported diabetes status: diabetes vs. no diabetes
- Sensitivity outcome: biomarker-confirmed diabetes (HbA1c $\geq$ 6.5% or fasting glucose $\geq$ 126 mg/dL)

Main predictors

- Age, sex, BMI, race/ethnicity, education, income ratio, hypertension, and smoking status

Data Source and Study Design

Data source

- National Health and Nutrition Examination Survey (NHANES), 2017–2018 cycle
- Publicly available U.S. population health dataset
- Combines questionnaire, examination, and laboratory data

Study design

- Cross-sectional observational study
- Adult participants aged 20 years and older
- Complete-case analysis after excluding missing or non-informative responses

Analytical approach

- Descriptive statistics, exploratory analysis, survey-weighted and unweighted logistic regression, sensitivity analyses, and model diagnostics

Cohort Definition

Step	N
NHANES 2017–2018 participants	9,254
Adults aged $\geq$ 20 years	5,569
Excluded: non-informative education responses	-13
Excluded: missing diabetes status	-175
Excluded: missing BMI	-380
Excluded: missing income ratio	-653
Excluded: missing hypertension status	-9
Final analytic cohort	**4,339**
Diabetes cases	701
No diabetes	3,638

$\frac{701}{4339} = 16.2\%$ (unweighted) Survey-weighted prevalence: 11.7%

Missing Data Assessment

Variable	Participants Excluded
Income-to-poverty ratio	653
BMI	380
Diabetes status	175
Education (refused/unknown)	13
Hypertension status	9
Smoking status	0
Age	0
Sex	0
Race/Ethnicity	0

Approach

- Complete-case analysis was performed.
- Participants with missing outcome or predictor information were excluded.
- Final analytic cohort consisted of 4,339 adults aged 20 years and older.

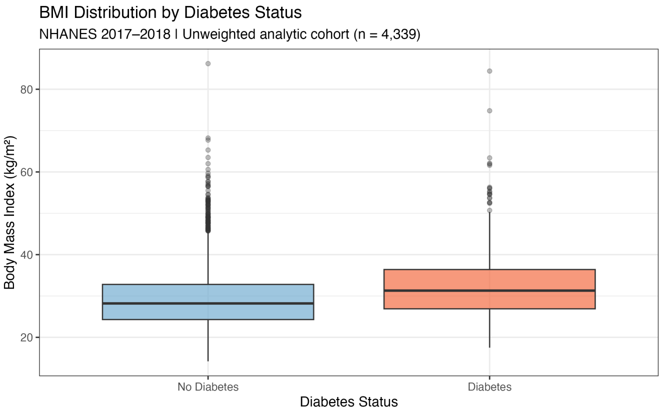
Descriptive Characteristics

Characteristic	No Diabetes	Diabetes
N	3,638	701
Mean age, years	48.6	63.5
Median age, years	48	65
Mean BMI	29.3	32.5
Median BMI	28.2	31.3
Mean income ratio	2.56	2.47

Initial observations

- Participants with diabetes were substantially older (mean 63.5 vs. 48.6 years).
- Participants with diabetes had higher average BMI (32.5 vs. 29.3).
- Income ratio showed only a small difference between groups.

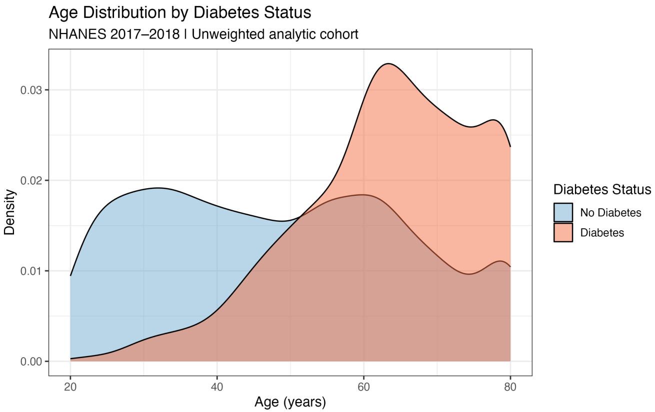
Exploratory Analysis: BMI



Interpretation

- BMI distribution was shifted upward among participants with diabetes.
- Median BMI was higher in the diabetes group (31.3 vs. 28.2).
- This supports BMI as a clinically relevant predictor for diabetes status.

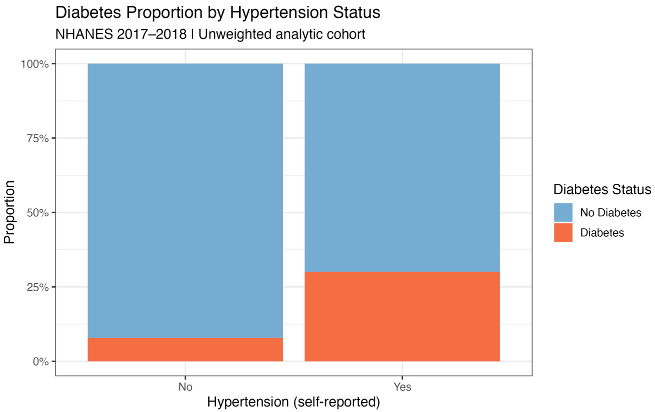
Exploratory Analysis: Age



Interpretation

- Diabetes was more concentrated among older adults.
- The no-diabetes group had a broader distribution across younger ages.
- Age was expected to remain important after adjustment.

Exploratory Analysis: Hypertension



Interpretation

- Diabetes prevalence was markedly higher among participants with hypertension.
- Approximately 30% of hypertensive participants had diabetes.
- Hypertension appeared to be one of the strongest clinical predictors.

Why Logistic Regression?

Study objective

- Identify factors associated with prevalent Type 2 diabetes.
- Estimate adjusted associations while controlling for confounding variables.

Why logistic regression?

- Outcome variable was binary (Diabetes: Yes/No).
- Provides interpretable odds ratios with confidence intervals.
- Widely used in epidemiology, biostatistics, and clinical research.
- Allows adjustment for multiple demographic and clinical covariates simultaneously.

Statistical Modeling

Three nested multivariable logistic regression models were estimated. The survey-weighted model (primary) used the NHANES complex sample design to produce nationally representative estimates.

$$\text{logit}(P(\text{Diabetes})) = \beta_0 + \beta_1(\text{Age}) + \beta_2(\text{Sex}) + \beta_3(\text{BMI}) + \cdots$$

Outcome

- Diabetes status: Yes/No

Adjusted covariates

- Age, sex, BMI, race/ethnicity, education, income ratio, hypertension, and smoking status

Survey weighting

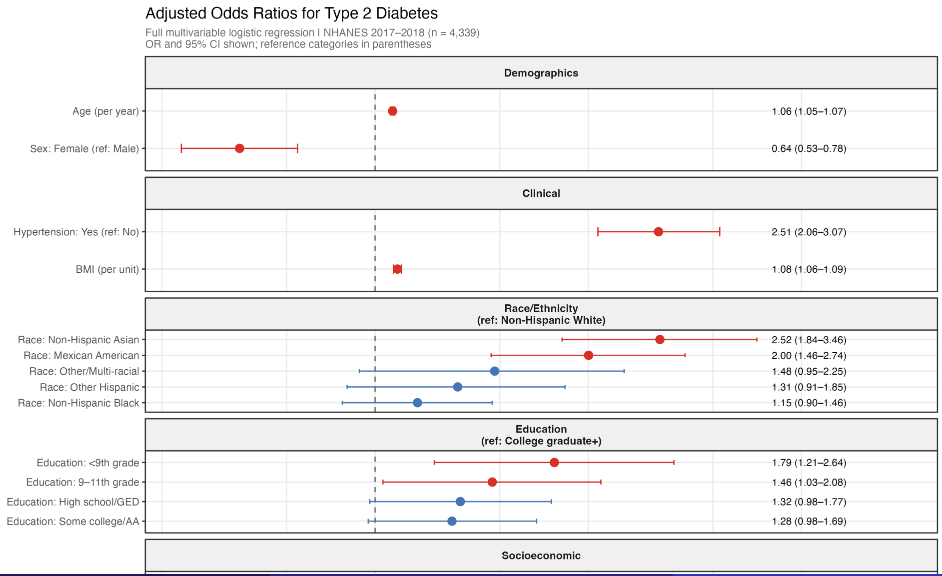
- NHANES MEC examination weights (`WTMEC2YR`) applied via `svyglm()` to account for the complex stratified sampling design

Main Regression Results

Predictor	Adjusted OR	95% CI	p-value
Unweighted model			
Age (per year)	1.06	1.05–1.07	<0.001
Female vs Male	0.64	0.53–0.78	<0.001
BMI (per unit)	1.08	1.06–1.09	<0.001
Hypertension vs No hypertension	2.51	2.06–3.07	<0.001
Ever smoker vs Never smoker	1.09	0.89–1.32	0.41
Income ratio	0.99	0.93–1.05	0.71
Survey-weighted model (primary)			
Age (per year)	1.06	1.05–1.07	<0.001
Female vs Male	0.66	0.45–0.96	0.031
BMI (per unit)	1.09	1.07–1.11	<0.001
Hypertension vs No hypertension	2.97	2.32–3.81	<0.001
Ever smoker vs Never smoker	1.11	0.80–1.55	0.54
Income ratio	1.02	0.93–1.13	0.65

Odds ratios adjusted for all covariates. Survey-weighted model accounts for NHANES complex sample

Adjusted Odds Ratios



Clinical Interpretation of Findings

Age

- Each additional year of age was associated with approximately 6% higher odds of diabetes (OR 1.06, 95% CI 1.05–1.07).
- The association remained stable across all model specifications and sensitivity analyses.

Body Mass Index

- Every one-unit increase in BMI corresponded to approximately 8% higher odds (OR 1.08, 95% CI 1.06–1.09).

Hypertension

- Participants with hypertension had 2.5-fold higher odds of diabetes in the unweighted model (OR 2.51); this strengthened to nearly 3-fold in the survey-weighted model (OR 2.97).
- This represented the strongest independent clinical association observed.

Race/Ethnicity

- Non-Hispanic Asian (OR 2.52) and Mexican American (OR 2.00) participants showed higher adjusted odds of diabetes compared with Non-Hispanic White participants

Sensitivity Analysis

Model specification sensitivity

Model	Covariates Included
Basic model	Age, sex, BMI
Clinical model	Age, sex, BMI, hypertension, smoking
Extended model	Age, sex, BMI, hypertension, smoking, race/ethnicity, education, income ratio

Biomarker-confirmed outcome sensitivity

- The primary analysis was replicated using a biomarker-confirmed diabetes outcome (HbA1c $\geq$ 6.5% or fasting glucose $\geq$ 126 mg/dL).
- Key associations for age, BMI, and hypertension were consistent in direction and magnitude across both outcome definitions.
- Non-Hispanic Black participants reached statistical significance in the biomarker model (OR 1.49, $p < 0.001$), likely reflecting known underdiagnosis in this group under self-report.

Overall

- Consistency of estimates across all sensitivity analyses supports the robustness of the

Multicollinearity

- VIF was low across all predictors.
- No evidence of problematic multicollinearity.

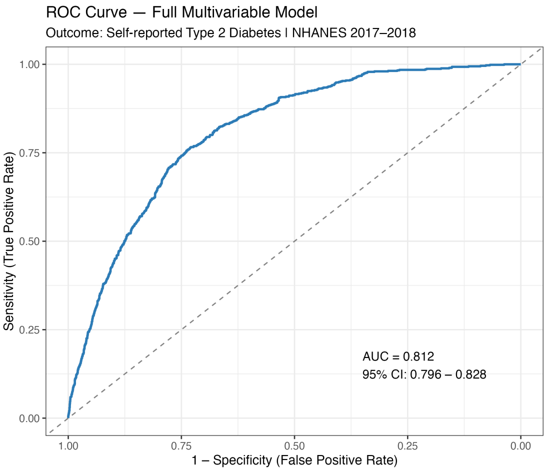
Calibration — Hosmer–Lemeshow

$$\chi^2 = 11.78, \quad df = 8, \quad p = 0.161$$

- No major evidence of poor model fit.

Discrimination — AUC

$$\text{AUC} = 0.812 \ \ (95\% \text{ CI} : 0.796\text{–}0.828)$$



Dashed line: chance-level discrimination (AUC =

Conclusions

Main findings

- Unweighted diabetes prevalence in the analytic cohort was 16.2%; the survey-weighted nationally representative estimate was 11.7%.
- Participants with diabetes were older (mean 63.5 vs. 48.6 years) and had higher BMI (mean 32.5 vs. 29.3).
- Older age, higher BMI, and hypertension were independently associated with increased odds of diabetes across all model specifications.
- Non-Hispanic Asian and Mexican American participants showed elevated adjusted odds relative to Non-Hispanic White participants.
- Hypertension demonstrated the strongest independent clinical association (unweighted OR 2.51, 95% CI 2.06–3.07; weighted OR 2.97, 95% CI 2.32–3.81).
- Findings were consistent across sensitivity analyses and between self-reported and biomarker-confirmed outcomes.

Overall conclusion

This analysis demonstrates how publicly available real-world health data, analysed with

Limitations

- Cross-sectional study design; associations cannot be interpreted as causal.
- Primary diabetes outcome was based on self-reported questionnaire data; a biomarker-confirmed sensitivity analysis was conducted as a validity check.
- Complete-case analysis was performed; if data are not missing completely at random, estimates may be subject to selection bias.
- The largest source of missing data was income-to-poverty ratio (653 participants excluded), which may disproportionately affect lower-income groups.
- Potential residual confounding remains from unmeasured variables not available in the dataset (e.g. physical activity, dietary patterns, family history).
- Results should be interpreted as an applied statistical portfolio analysis rather than a definitive epidemiological estimate.

Methods and Tools

Software and packages

- R
- `nhanesA` — data download
- `tidyverse` — data wrangling
- `survey` — weighted regression
- `gtsummary` — summary tables
- `broom` — tidy model output
- `ggplot2` — visualisation
- `pROC` — AUC and ROC curve
- `car` — VIF diagnostics
- `ResourceSelection` — H-L test

Statistical methods

- Descriptive statistics (Table 1)
- Exploratory data analysis
- Univariable logistic regression
- Multivariable logistic regression (unweighted)
- Survey-weighted logistic regression (`svyglm`)
- Model specification sensitivity analysis
- Biomarker outcome sensitivity analysis

Model evaluation

- Variance Inflation Factor (VIF)
- Hosmer–Lemeshow goodness-of-fit test
- AUC / ROC curve

Thank You

Questions?

Appendix: Full Multivariable Regression Results (Unweighted)

Predictor	OR	95% CI	p-value
Age	1.06	1.05–1.07	<0.001
Female vs Male	0.64	0.53–0.78	<0.001
BMI	1.08	1.06–1.09	<0.001
Mexican American vs Non-Hispanic White	2.00	1.46–2.74	<0.001
Other Hispanic vs Non-Hispanic White	1.31	0.91–1.85	0.14
Non-Hispanic Black vs Non-Hispanic White	1.15	0.90–1.46	0.27
Non-Hispanic Asian vs Non-Hispanic White	2.52	1.84–3.46	<0.001
Other Race vs Non-Hispanic White	1.48	0.95–2.25	0.076
Less than 9th grade vs College graduate+	1.79	1.21–2.64	0.003
9–11th grade vs College graduate+	1.46	1.03–2.08	0.035
High school graduate vs College graduate+	1.32	0.98–1.77	0.066
Some college/AA vs College graduate+	1.28	0.98–1.69	0.073
Income ratio	0.99	0.93–1.05	0.71
Hypertension vs No hypertension	2.51	2.06–3.07	<0.001
Ever smoker vs Never smoker	1.09	0.89–1.32	0.41